Moca User & template lifecycle automation

Self-service workflows with policy-based gates and provenance tracking

2026-11-12

Outline

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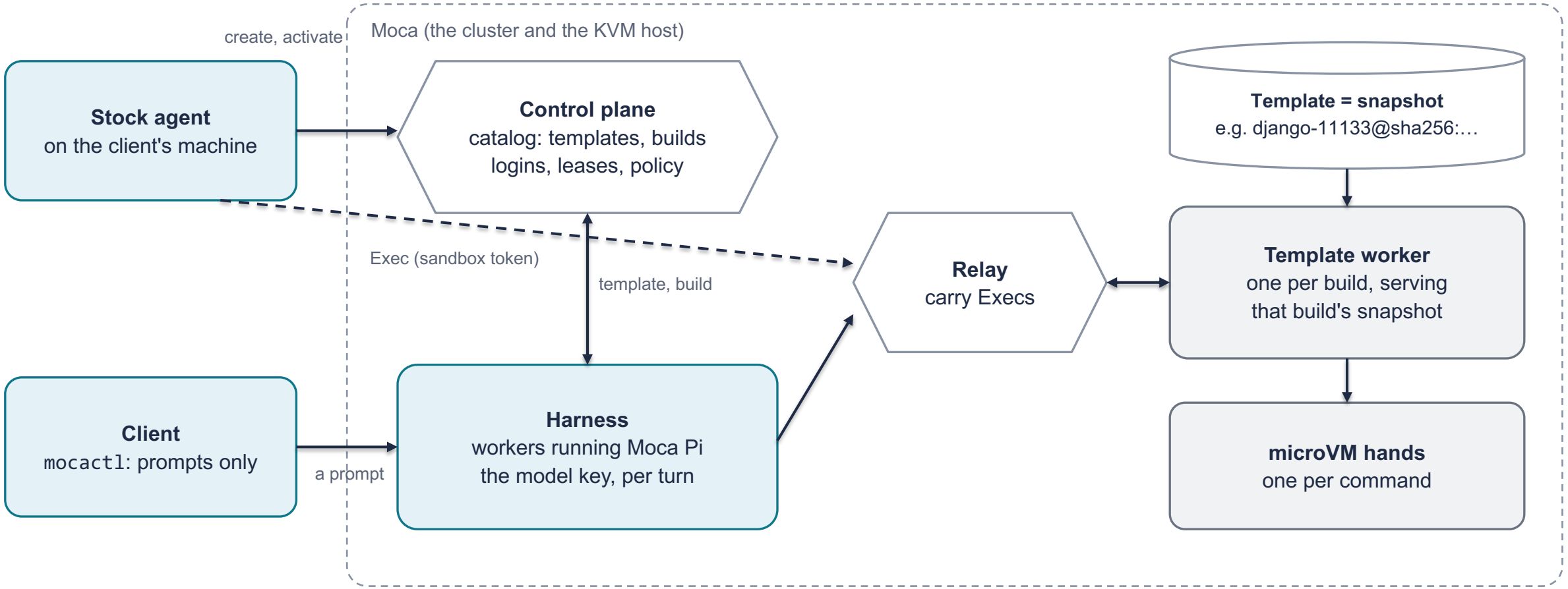
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The brain thinks, the hands act; a template shapes the hands

Moca Pi brain, or a stock agent, with Moca microVM sandbox hands



The hands never hold the model key or the agent
they get commands and return output; a template decides what the hands contain

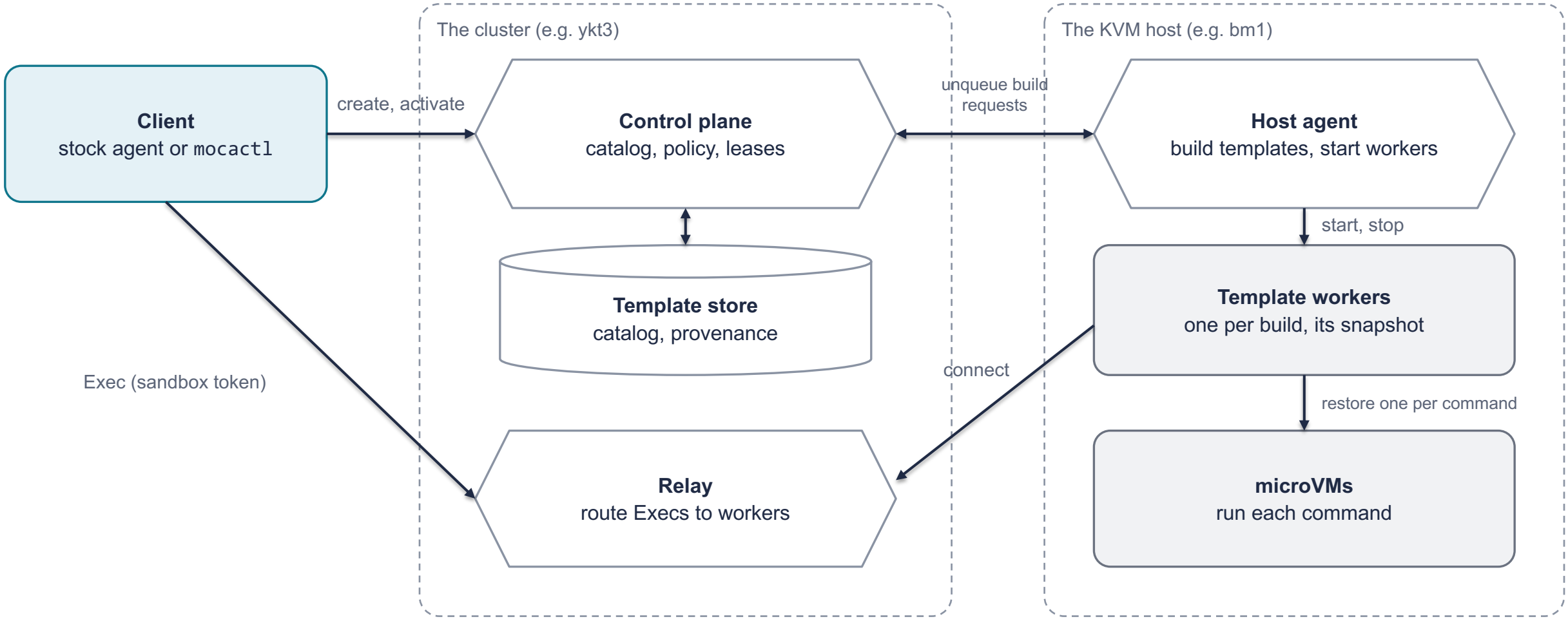
Self-serve: policy, not a person, decides; every step is recorded

The design in five ideas (template spec §3, §6-§8; enrollment spec §3)

Templates are declared in a store	a name over immutable builds, each named by its snapshot's manifest hash
Policy gates every image	rules checked at create; a build becomes `registered` by itself once its probe passes in a restored microVM and the rules hold again
Hosts build and run; the control plane decides	a host agent claims builds and starts workers; clients activate by lease
Users and roles through the API	the settings bootstrap the first admins; enroll, grant and revoke are API calls, evaluated per request
Every step is a chained record	append-only provenance named by digest, verifiable offline, off the cluster

One control plane decides; hosts build and run; clients only ask

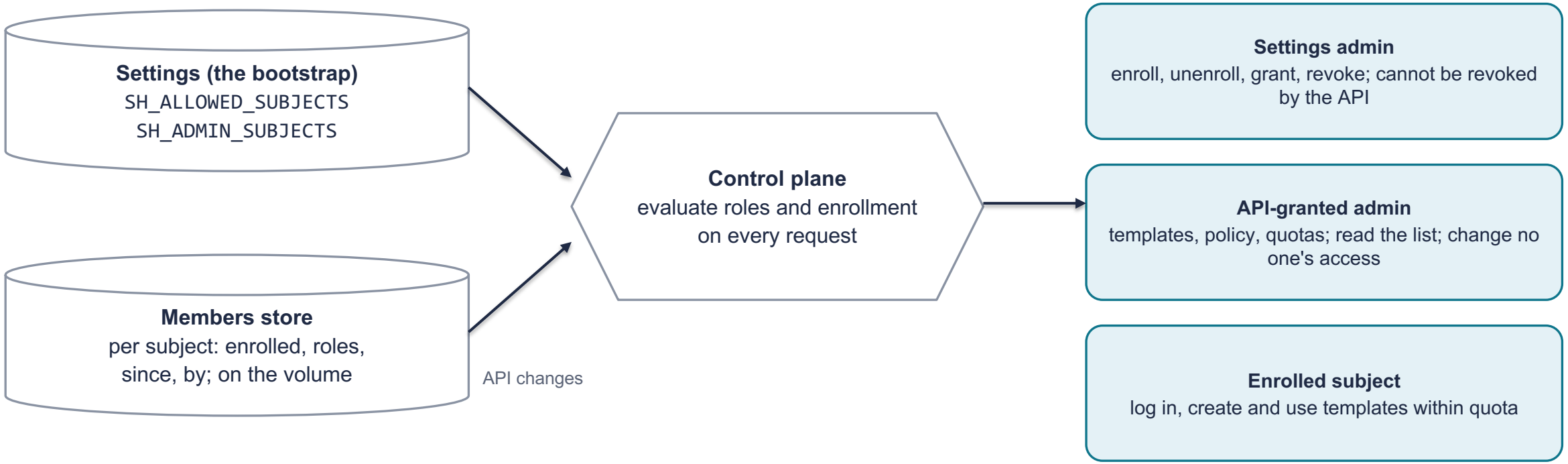
Who runs where: the cluster (OpenShift; e.g. ykt3) and the KVM host (bare metal; e.g. bm1)



A worker **connects** to the relay and holds one two-way gRPC stream (Attach): commands come down it, results go back up. Readiness is a worker connected to the relay; the cluster holds no VM, the KVM host no policy.

The settings name the first admins; the API manages everyone else

Who may use the service, and who is admin (enrollment spec §3, §6)



The recovery path stays the deployment's
the settings' admins are always admins: none can be revoked through the API, and none can revoke itself

Demo: a user's lifecycle, in seven commands

As the demo runs it: the admin on one laptop, the user on another (demo runbook §1, §5)

Who	Command	What the room sees
user	`mocactl sbx login`	GitHub's device code; `403 not_enrolled` until step 2
admin	`sbx admin subjects enroll github:<id>`	a `member` record; the user's next request works
admin	`sbx admin subjects`	each subject, its roles, and `env` or `api` as source
admin	`sbx admin subjects grant github:<id> --until 30m`	the user's next request is admin
admin	`sbx templates quota github:<id> --templates 10 …`	old → new; the user's `sbx quota` shows it
admin	`sbx admin subjects revoke github:<id>`	admin gone at the next request
admin	`sbx admin subjects unenroll github:<id>`	the user's next request: `403`

No settings edit, no setup.sh, no redeploy
every step is an API call that takes effect at the next request and leaves a record

Demo: a template's lifecycle, image to retire

As the demo runs it, as one enrolled user, profile `base` (demo runbook §1-§4)

Step	Command	What the room sees
1 what I may do	`sbx quota`, `sbx policy`, `sbx hosts`	limits and use, profiles, host slots
2 check, then create	`sbx templates create --spec F --dry-run`, then without	local checks pass; a build queued
3 follow the build	`sbx templates follow ID`	queued → building → probing → registered
4 versions	`templates build`, `tag`, `edit`	a second build; a tag names one
5 use	`templates activate --wait`; `sbx create --template ID`; `sbx run`	a lease; a sandbox on that build
6 release, retire	`templates deactivate`; `templates retire --wait`	the worker stops; a later create: `410`

Two refusals, shown between steps 2 and 3; nothing is built

1. Refused by mocactl, before sending
`create --spec F --image <another digest>`
rule `M7.spec_image`: `--image` supplies an image a spec lacks, never a different one; refused locally, nothing sent

2. Refused by the control plane's policy
`create --spec F --profile benchmark`
`422 policy_refused`: no trusted gate record; a `check` record

Every step leaves a record on the template's chain: create, gate, build, check, activate, retire.

Five routes, one mocactl command; each change is a record

Reads: any admin. Changes: an admin named in the settings (enrollment spec §4)

Change	Route	`mocactl sbx admin subjects …`	Record
list subjects, roles, source	`GET /v1/subjects`	(no argument)	none
enroll	`PUT /v1/subjects/{s}`	`enroll SUBJECT`	`member`
unenroll	`DELETE /v1/subjects/{s}`	`unenroll SUBJECT`	`member`
grant admin, optional expiry	`PUT …/roles/admin`	`grant SUBJECT --until T`	`role`
revoke admin	`DELETE …/roles/admin`	`revoke SUBJECT`	`role`

Refusals: `409 bootstrap_subject` for a subject the settings name; `409 admin_granted` for an unenroll while an admin grant keeps the subject enrolled; `403` for a non-admin.

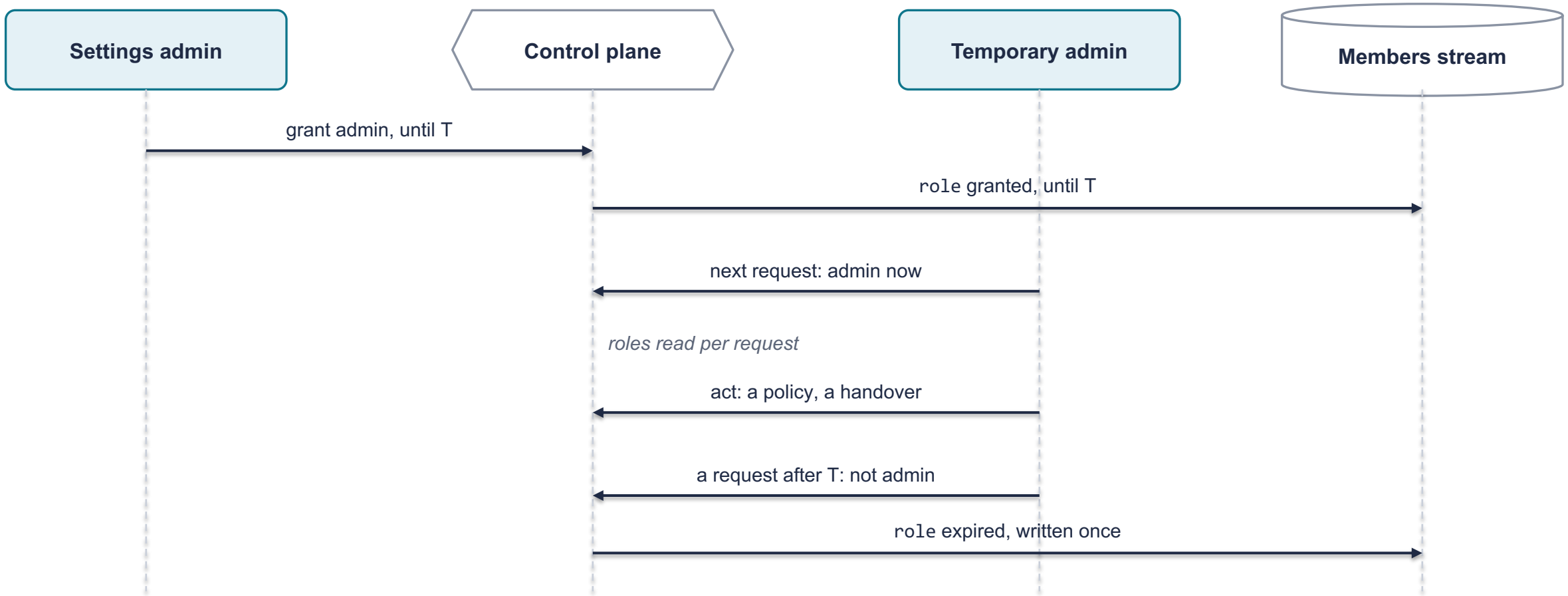
The client checks its target: a login for another control plane is refused before any write; each write prints old → new.

Subjects are login ids

`github:<numeric id>` (or `oidc:<sub>`), from the existing login; never a name

A grant takes effect at the next request, and can expire by itself

A temporary admin for one task, with no setup.sh (enrollment spec §5)



No revoke step and no redeploy: the grant lapses on its own

Onboarding a user: one GitHub login, one enroll

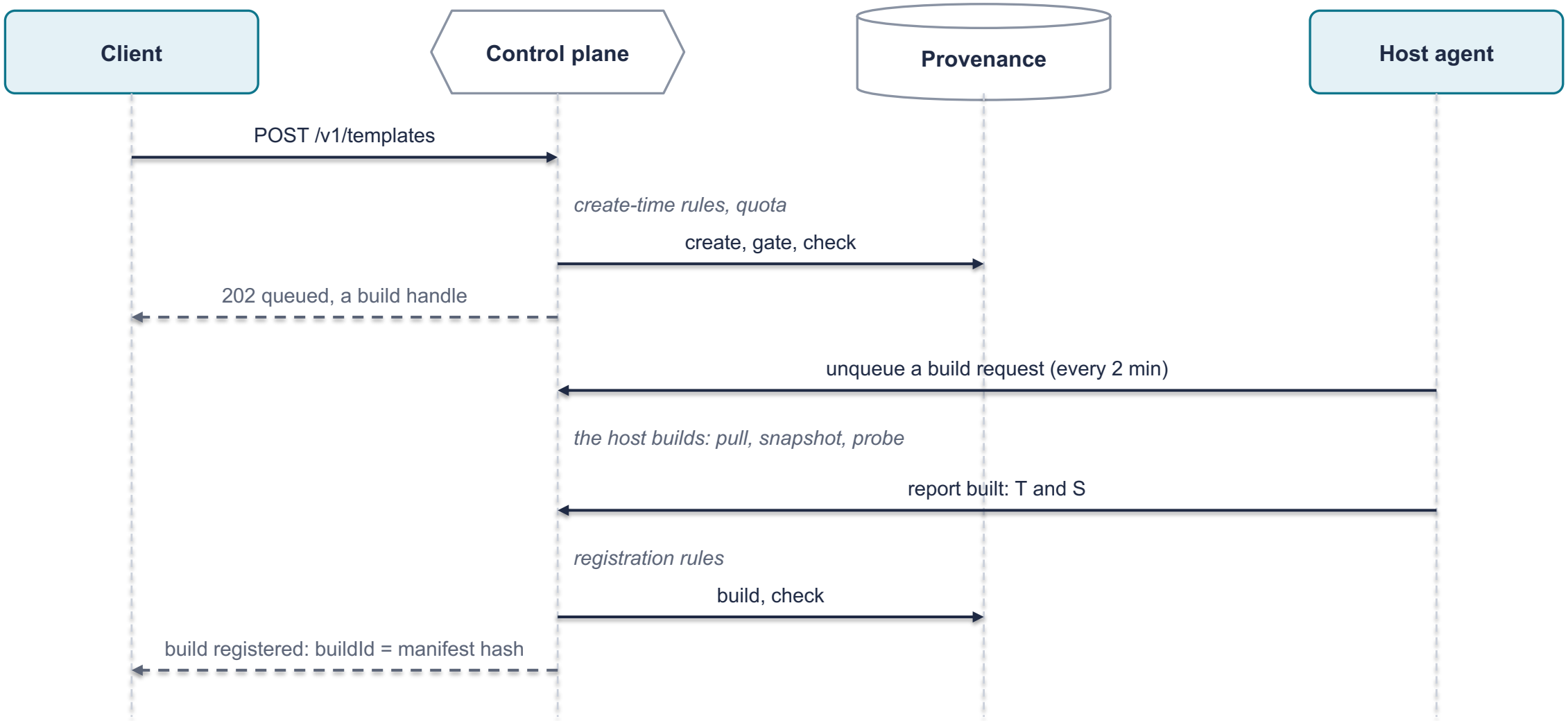
Moca's own device-flow login; no token is pasted, no settings edit

1	**Get a GitHub account**	the subject is its numeric id, `github:<id>`
2	**`mocactl sbx login`**	GitHub's device flow, through the deployment's OAuth app; `403 not_enrolled` until enrolled
3	**An admin enrolls it**	`sbx admin subjects enroll github:<id>`; a `member` record names who did it
4	**The next request works**	`sbx templates`, `sbx quota`, `sbx create`; no redeploy

One login per control plane
each namespace is its own deployment: a user logs in to each one it uses, and is enrolled in each

Create to registered: no approver, every decision recorded

Sequence 1: onboarding (spec §4, §6.1, §8)



Refusals are answers with a rule, not silence

What a client sees when a create cannot proceed (spec §4, §8; G21, G22, G29)

Case	Answer	Recorded
Policy refuses the create	422 `policy_refused` with the rule id	a `check` record
Another subject owns the id	409 `template_exists`	nothing queued
Same inputs as a build a host still holds	200 or 202 with `reused: true`	no second build
Over a quota	429 `quota_exceeded`, naming which	nothing queued
The owner withdraws a build	cancel: `failed` at step `cancelled`	a `build` record
The build or probe fails on the host	`failed`, with step and code (`probe_failed`)	a `build` record

Nothing that names where an image lives leaves the host
failure text is redacted; a record naming the registry is not sent at all (G25, G28)

A create names a profile; the profile names the rules

Policy profiles an admin sets (template spec §8; subject-trust spec §3)

Profile	Gate record	Egress	For
`base`	none	per spec	general images, no benchmark claim
`benchmark`	trusted by an admin (X1)	none (X11)	curated benchmark tasks
`benchmark-self`	the creator's own (X1, self)	none (X11)	a user's own benchmark tasks
`general-self`	the creator's own (X1, self)	the host's allowlist	general users who need a network

Checked at create and at registration: the id; the image, by digest, from an allowed registry and repository; the spec; the size envelope; each rule the profile names.

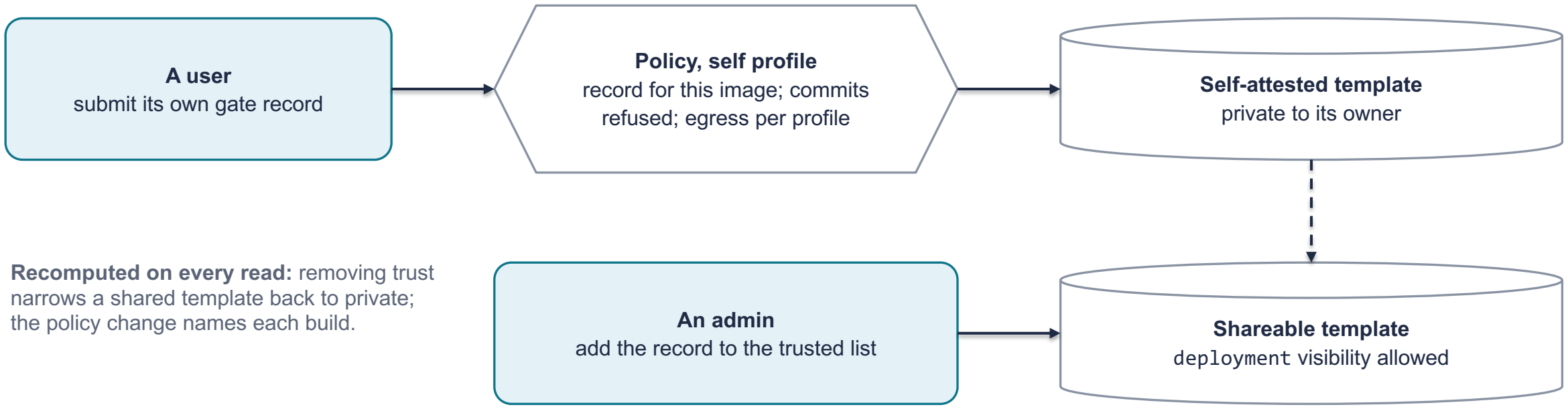
Quotas per subject: templates owned, builds queued, templates active at once; `mocactl sbx quota` shows limit and use.

A refusal names its rule

`422 policy_refused` with the rule id, and a `check` record on the caller's own template's stream

Your own gate record counts for your own templates, privately

Trust comes from who submits a record, never from what it says (subject-trust spec §2-§6)



Asking for `deployment` visibility before trust
is refused `409 self_attested`, on the create, on a new build, and on an edit

The client's whole path, in mocactl

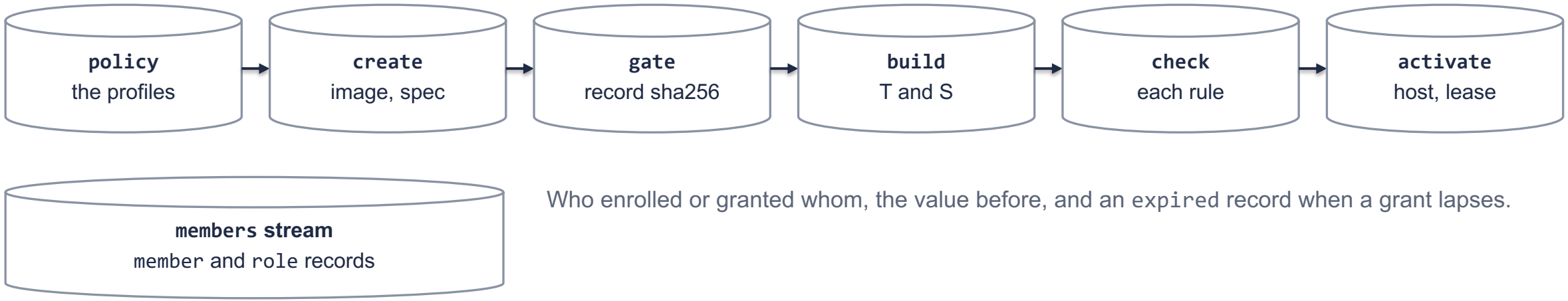
Each step a command, `--json` on all of them (mocactl template commands spec §3, §4a)

Step	Command	What it does
before a create	`sbx quota`, `sbx policy`	limits and use; the profiles and rules
check locally	`sbx templates create --dry-run`	the rules the inputs decide; creates nothing
create and follow	`sbx templates create`, `follow`	post the spec and gate record; wait for registered
new build, cancel	`sbx templates build`, `cancel`	a build of an existing template
activate, use	`sbx templates activate --wait`, `sbx create --seed`	lease a slot; seed the workspace
end of life	`sbx templates retire --wait`	until no host holds the build
users and admin	`sbx admin subjects …`	print old → new; detect a concurrent write
a record per call	`--record F`	one line per HTTP attempt, no secret in it

Errors keep the server's reason
the rule id, the quota hit, the readiness and the queue position; exit 5 for a wait that timed out

Every step is a chained record, checkable offline

Provenance (template spec §7; enrollment spec §8): digests, not signatures



- **One file per record**, named by the sha256 of its canonical JSON; each names its predecessor. Any edit, drop or reorder breaks the chain.
- **Verified offline** by a stdlib Python script or the same rules in TypeScript, from an export, on a machine with no access to the cluster.

An export proves itself consistent, not complete
compare each stream's head with a copy kept elsewhere to catch a dropped tail

A build has two states: how far from usable, and whether it runs now

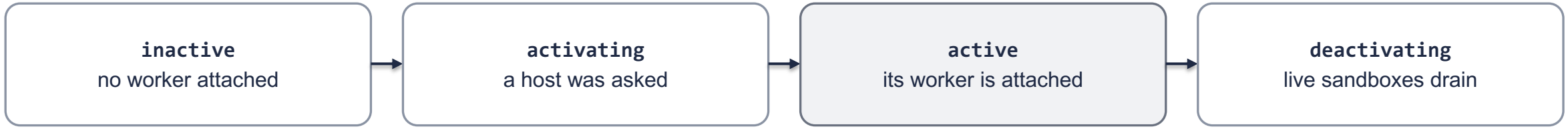
Onboarding is set by the host's reports and policy; readiness by what is attached

Onboarding (spec §3.3)



Off the path: `failed` (a build or probe failure, with its step; or the owner's cancel), `rejected` (policy at registration, with its rule), `retired`.

Readiness of a registered build

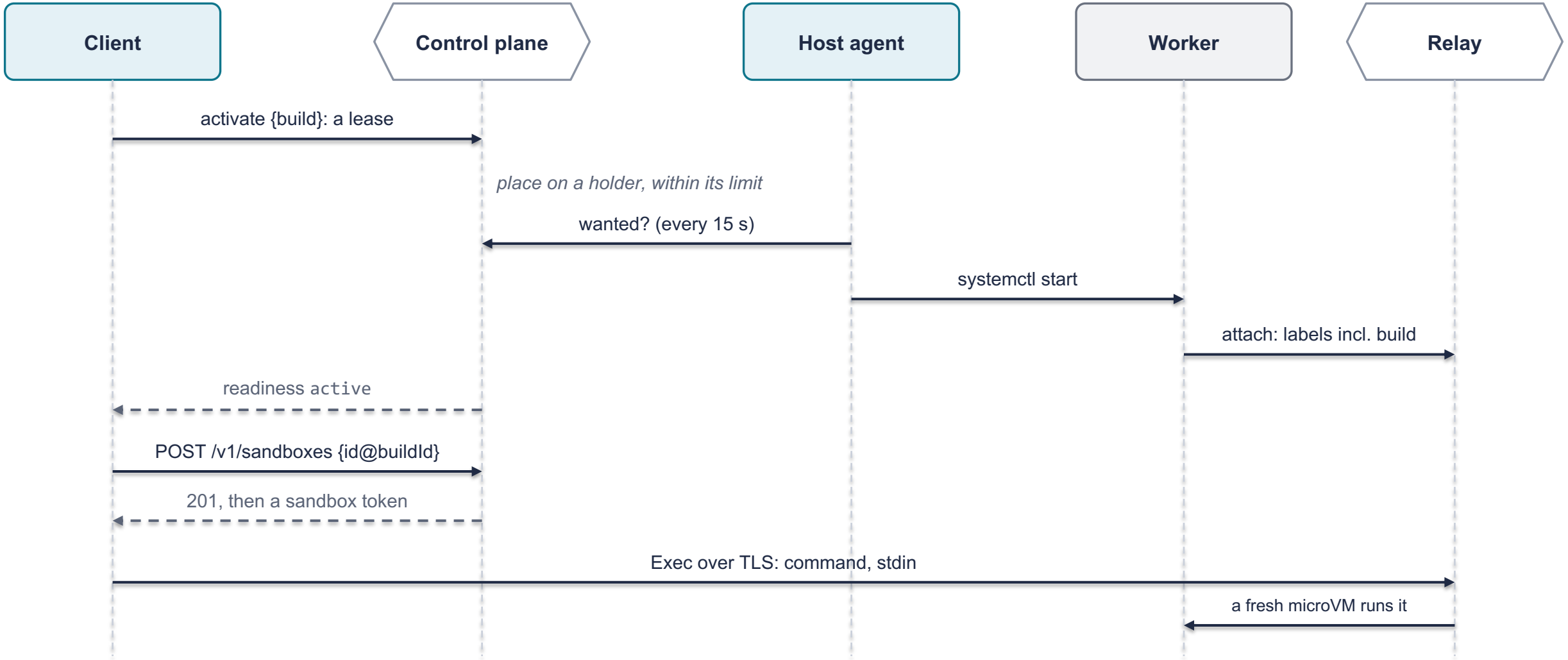


`unavailable`: no host holds its snapshot, or every holder is full. `active` is derived from the relay's attached record, so the state cannot claim a worker that is not there.

A build, in Moca
the KVM host turns the template's container image into a microVM snapshot, then runs the spec's probe in a VM restored from it

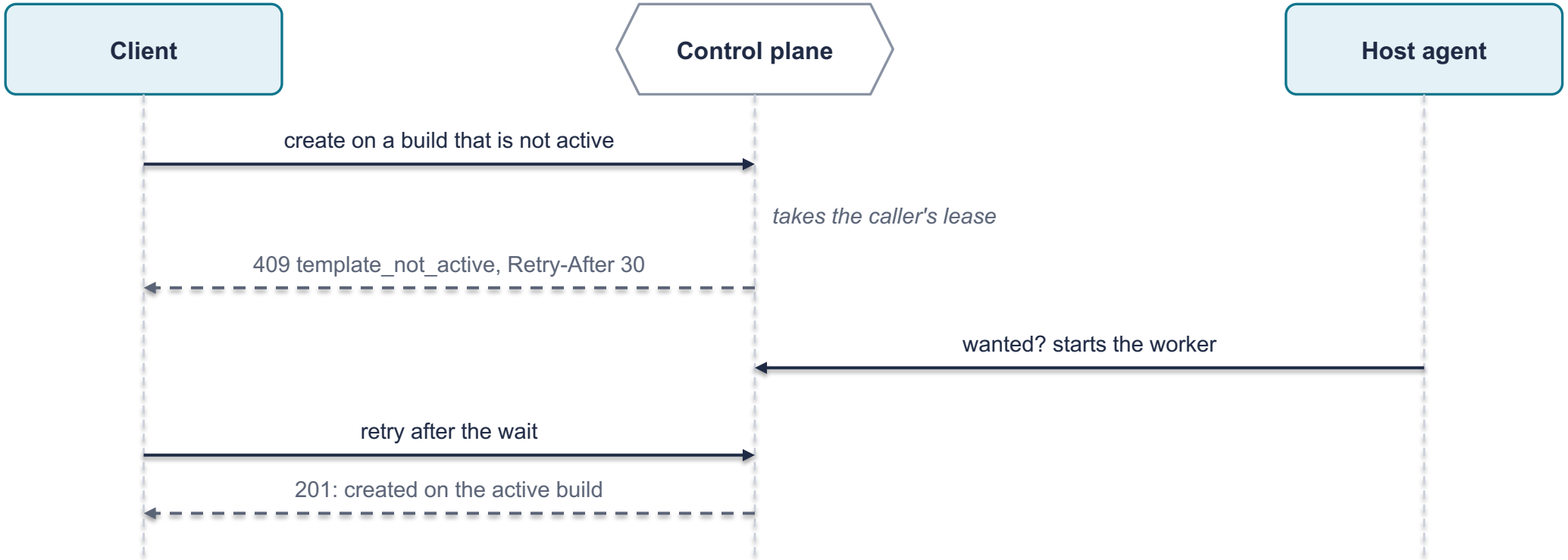
Activation is a lease; the worker starts in seconds

Sequence 2: activate, then a sandbox on the pinned build (spec §6.3; G31)



First use: name a template and go

Activation on first use (spec §6.4), for sandboxes and sessions alike

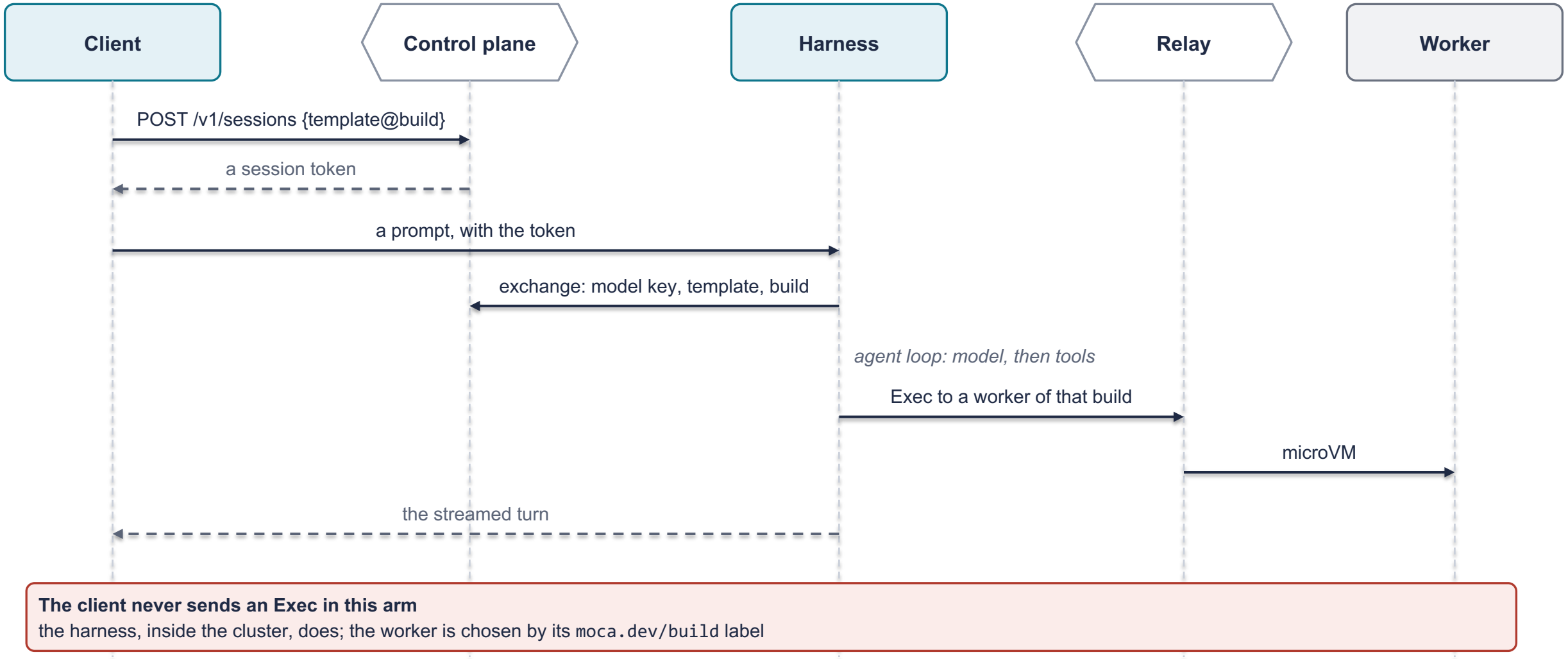


`activate: false` starts nothing. No sandbox is kept while waiting, so a wait never uses the caller's sandbox cap. `mocactl` retries for up to 10 minutes, saying why.

Without host agents nothing changes: the create succeeds as before

A Moca Pi session runs only on its build

Sequence 3: a session pinned to a build (combined control plane §3.4; G19)



One control plane serves both kinds of brain

The combined route profile (combined control plane design, C1-C4)



Decision	Chosen
C1 How	a third profile, `combined`; `harness` and `sandbox` unchanged
C2 Login	open as on `harness`; enrollment checked on sandbox routes and template writes
C3 Where	a new namespace, so the validated SBX0 stays as it is
C4 Turns by build	in: a worker label and a filter (G19)

The API covers the whole lifecycle, create to retire

Each activity a route and a mocactl subcommand; each writes a record naming who

Activity	Who	Record it writes
create a template, or a new build	any enrolled client	create, gate, check
build, probe, register by policy	the host agent, then policy	build, check
activate, deactivate (a lease)	the owner; anyone, if shared	activate, deactivate
move a tag onto a new build	the owner	tag (new and old build)
edit description, visibility	the owner	visibility (new and old)
retire a template or a build	the owner	retire (and the host's removal)
reject a build after the fact	an admin, with a reason	reject
transfer ownership	an admin	owner
set quotas, default or per user	an admin	policy

Deployed and shown end to end
a retire never cuts off running work: refused while a sandbox or session uses the build

Four windows validated the lifecycle end to end

Spec §10 steps, as run 2026-11-04 to 2026-11-05 (no model)

Step	Window 1	Window 2	Windows 3-4
Create, build, register	all four	all four, fresh	all four, each
Refused create (X10)	yes	yes	yes
Broken probe	not reached	`probe_failed`	step `probe` (w4)
First use	not built	409, then active	as w2
Served build = pinned	every sandbox	every sandbox	every sandbox
Patches match recordings	not compared	every patch	every patch
Offline verification	83 records	140 records, all chains verified	not repeated

Step 3 (a second client on the same instance) passed separately with a second enrolled subject: its create and its new build `409`, its cancel `403`.

Create to retire, through the API alone, on the cluster

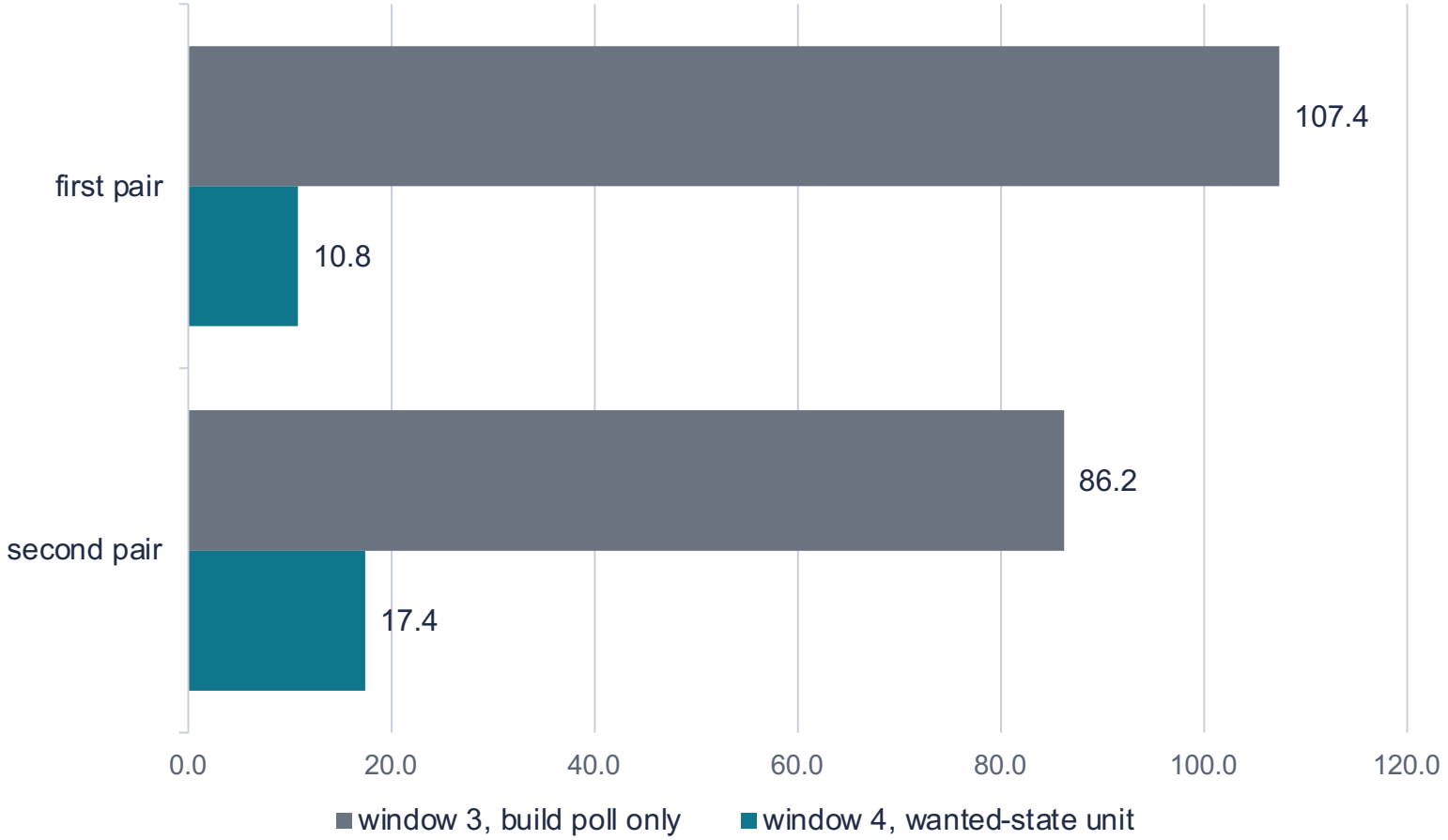
The lifecycle window, 2026-11-06: a user with no role; no operator step (no model)

Stage	What happened
create, build, register	all four by policy; the bad gate refused (X10); the bad probe failed
activate, first use, use	a lease in 23-26 s; every sandbox on its pinned build
retire	every build; a later create refused `410`; readiness `retired` in ~1 s
the host removes it	the KVM host's agent, by itself; `heldBy` emptied in 17-32 s
provenance	every chain before is an unchanged prefix after; 84 records appended
composition	an independent composer: every link of each rep unique

A whole-template retire is every build
earlier windows' builds of those templates were retired too: a new build is the way back

Activation fell from ~100 s to ~11-17 s

Seconds from each activation's request to `active`, both builds of a pair together (G31)



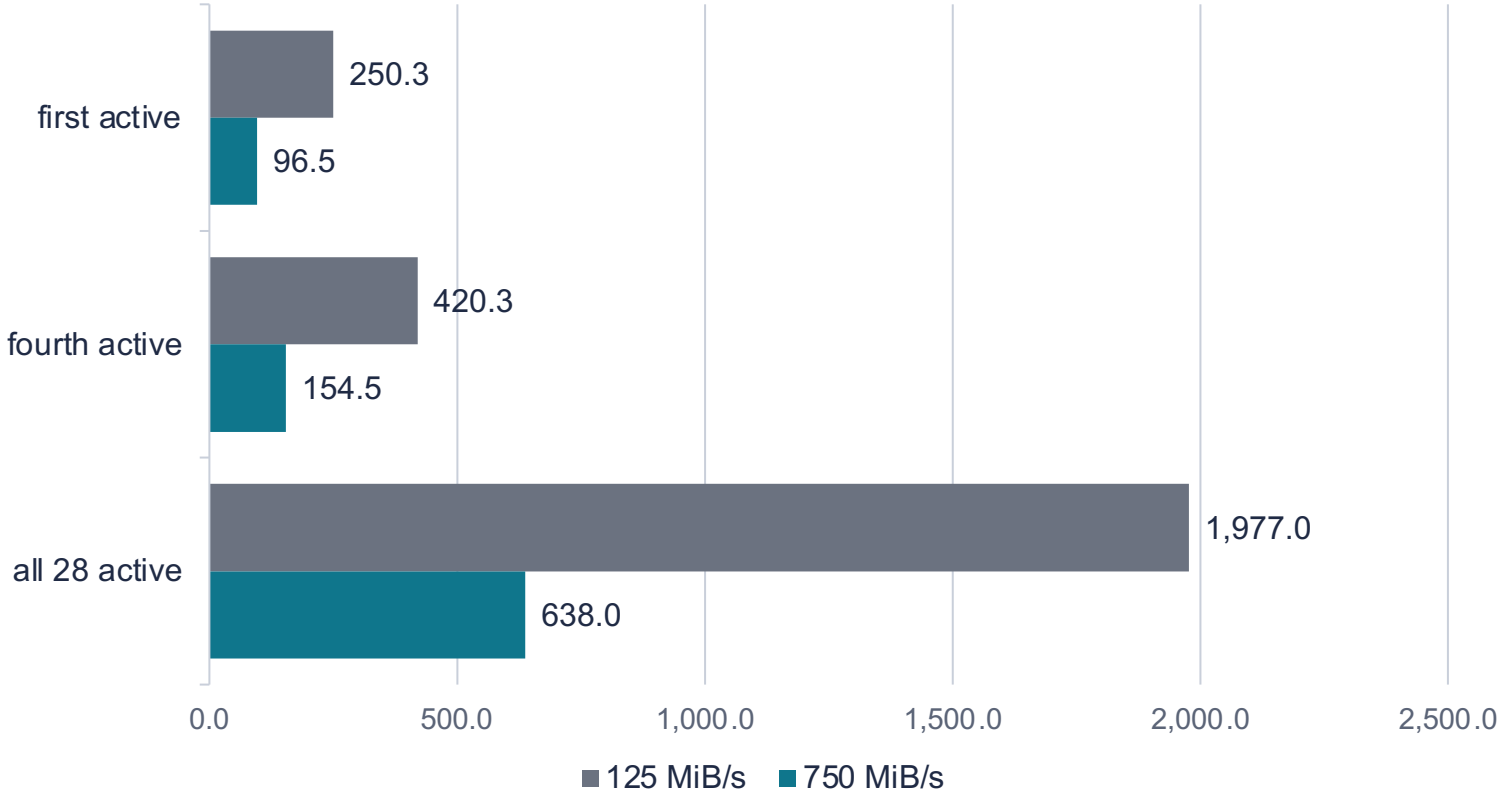
Before: an activation waited for the build poll, which starts 2 minutes after the previous one ends, and behind any build.

After: a separate unit applies the wanted state every 15 s; a lock keeps it and the build poll from acting at once.

One pair per window
a measurement, not a distribution

Disk throughput sets the cold start: 3.1× faster

Seconds until active, 28 activations posted at once, same builds; only the volume's throughput changed (I28b-7-d, D1 and D1b, 2026-11-11)



Mean disk read: 116.1 → 364.7 MiB/s over the round.

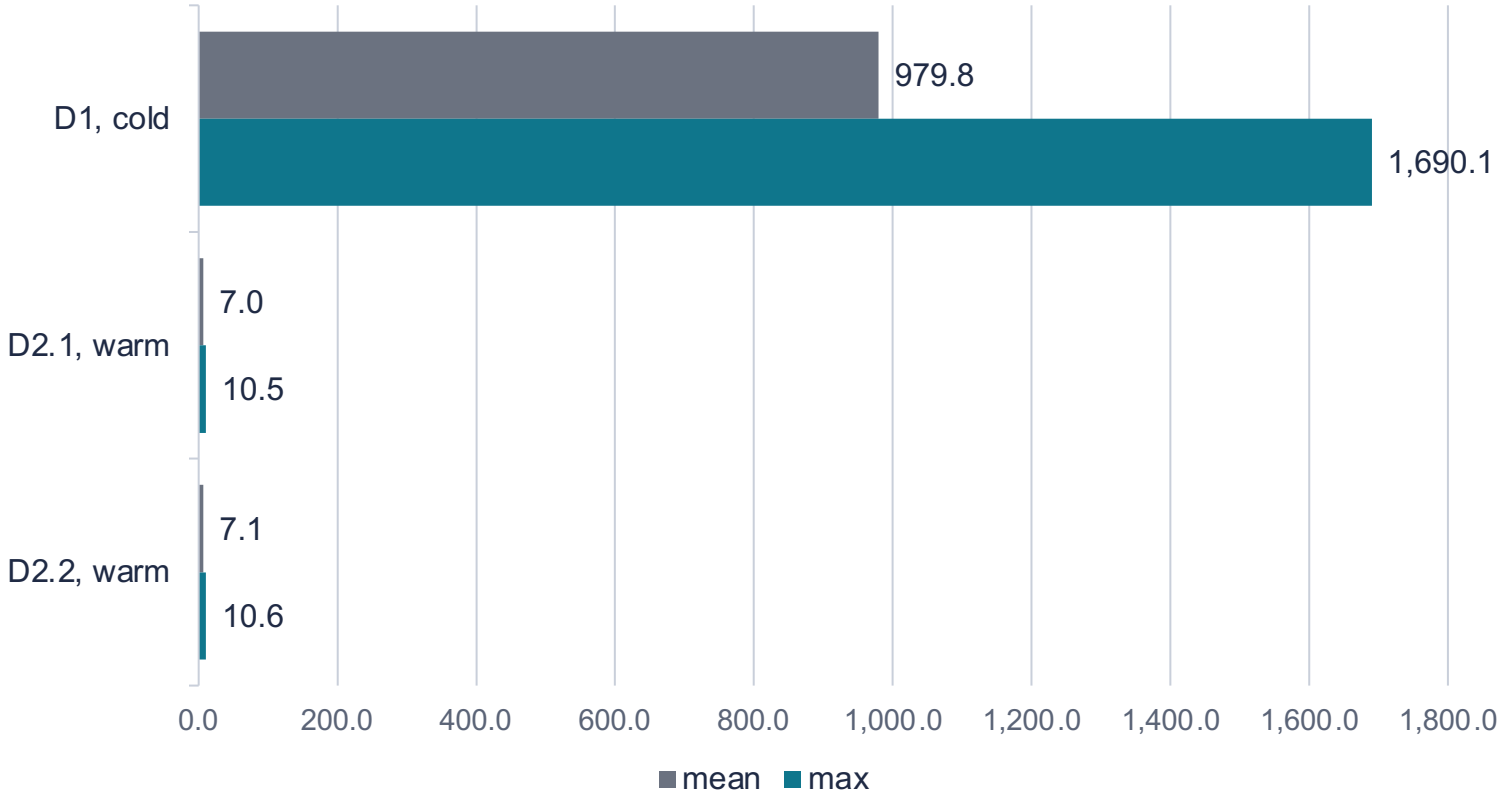
Time per read: 6.99 → 1.65 ms, at 3,000 IOPS both times.

Each cold start read about its whole snapshot
a mean of 8.70 GiB per start at 750 MiB/s: reading less per start is the next lever

one KVM host, 28 SWE-bench Lite templates (4 GiB, 2 vCPUs, snapshots 7.3-13.6 GiB), start limit 4, page cache dropped before each cold round

A warm worker activates in seconds: 7 s, not 980 s

Mean seconds per activation, 28 posted at once, three rounds back to back (l28b-6-d, 2026-11-10/11, at 125 MiB/s)



Warm by the idle grace (900 s): a released worker stays attached, so the next activation needs no restore from disk.

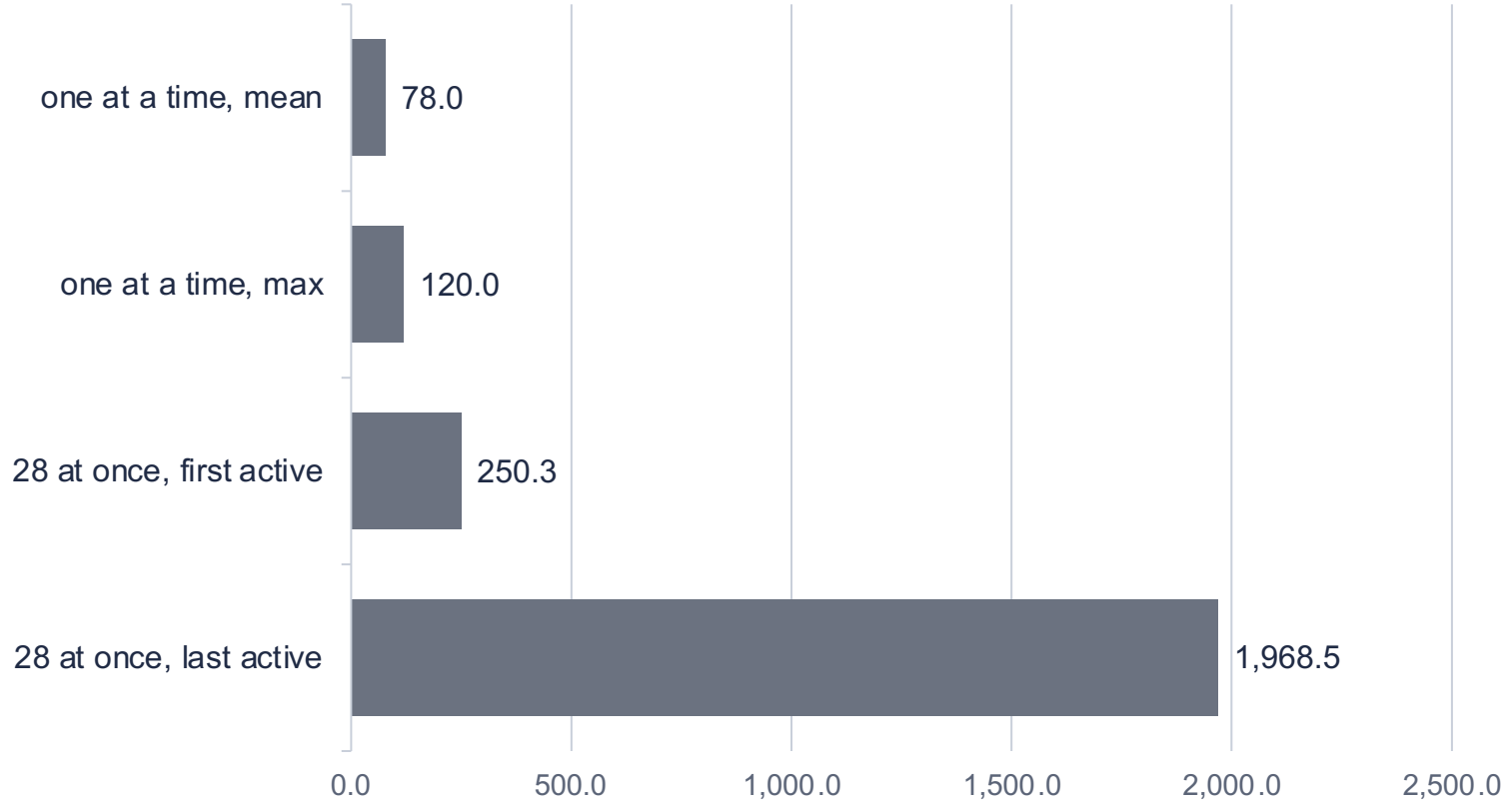
All 28 done after: 1,809 s cold; 19 s in each warm round.

D1's mean covers 23 activations
the client stopped waiting on 5, which the control plane completed; in both warm rounds all 28 workers were still attached

one KVM host, 28 SWE-bench Lite templates (4 GiB, 2 vCPUs, snapshots 7.3-13.6 GiB), start limit 4, page cache dropped before each cold round

28 cold starts at once queue behind the start limit

Seconds until active; alone, a cold start takes about 78 s (125 MiB/s, start limit 4; l28b-3-d serial-1, 2026-11-09/10; l28b-7-d D1, 2026-11-11)



Four start at a time: the rest wait, so the last of 28 is active after about 33 minutes.

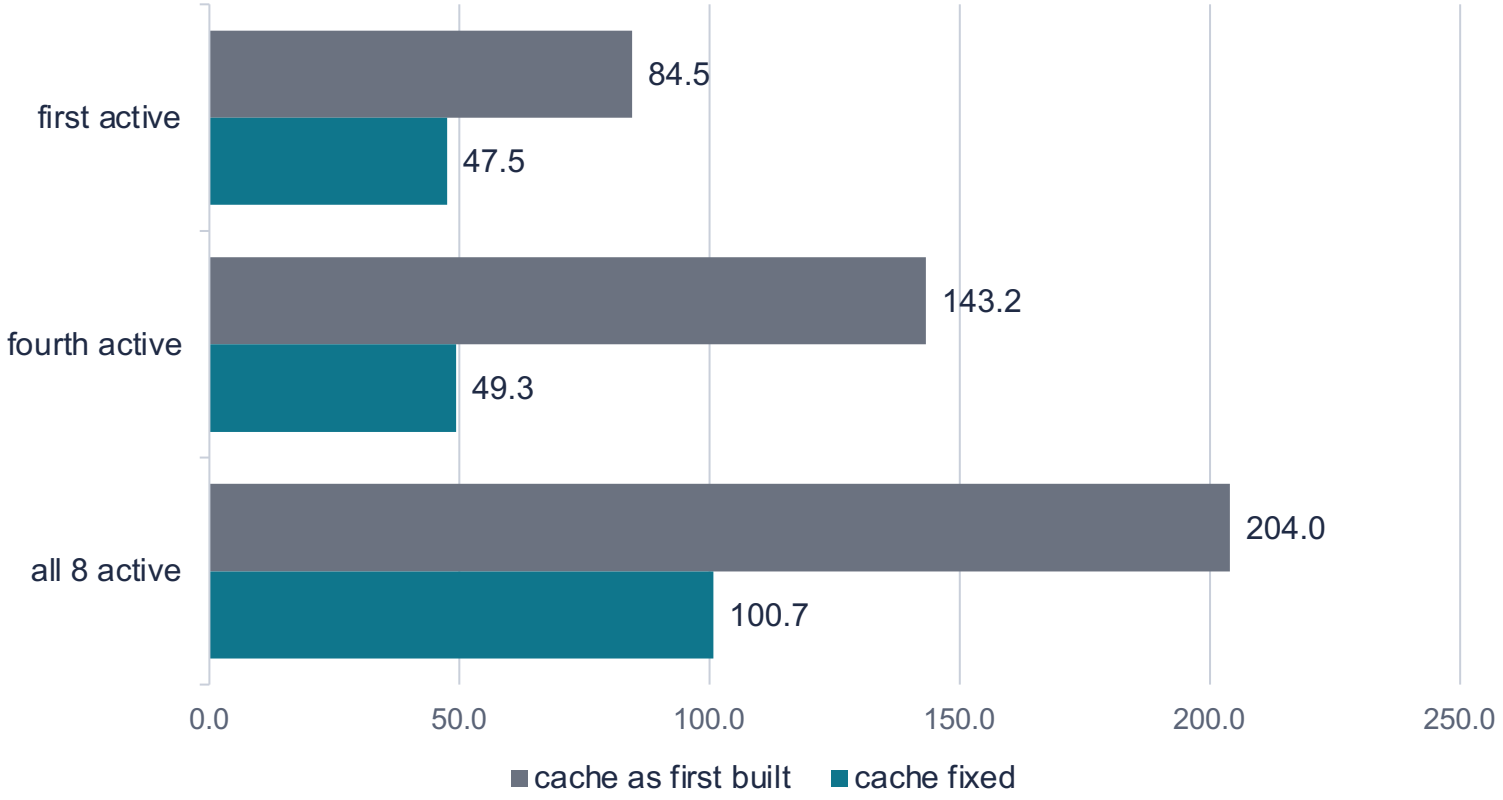
Then the disk: four starts at once share one volume's throughput (previous charts).

One pass per condition
a measurement, not a distribution

one KVM host, 28 SWE-bench Lite templates (4 GiB, 2 vCPUs, snapshots 7.3-13.6 GiB), start limit 4, page cache dropped before each cold round

The fixed verify cache halves a cold start's reads

Seconds until active, 8 cold activations posted at once (G50 window, 2026-11-12, 750 MiB/s, start limit 4)



Read per start: 8.99 → 4.00 GiB, the memory file alone, which a start still pins.

Files re-hashed per start: 4 of 4 → 0 of 4.

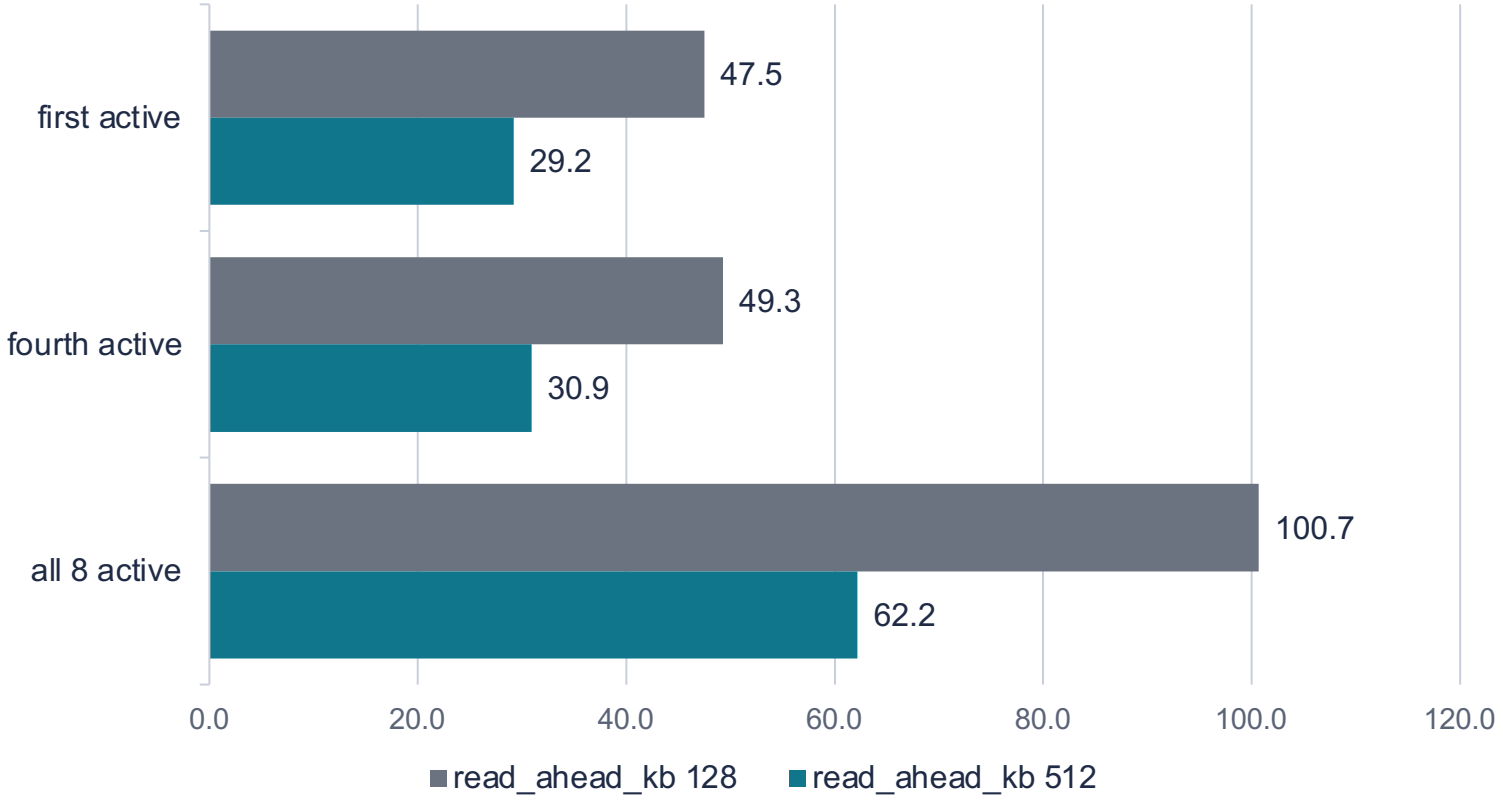
Why the first build missed

each VM restore links and chowns the files into its jail, which moved the ctime the cache keyed on

one KVM host, 8 SWE-bench Lite templates (4 GiB, snapshots 7.3-13.5 GiB), page cache dropped before each round

Read-ahead 512 lets the disk reach its cap

Seconds until active, 8 cold activations at once, fixed verify cache; only read_ahead_kb changed: 1.6× faster (G50, 2026-11-12)



Disk, busy intervals: 420.8 → 742.3 MiB/s median, at the volume's 750 MiB/s cap.

Same work: about 4 GiB read per start; reads stayed 127.5 KiB, the device's largest.

One round per setting
a measurement, not a distribution

one KVM host, 8 SWE-bench Lite templates (4 GiB, snapshots 7.3-13.5 GiB), page cache dropped before each round

SWE-bench Lite on templates, in groups, retire between

Lite-28 on templates, 2026-11-06/07: stock brains, the sandbox profile, a real model

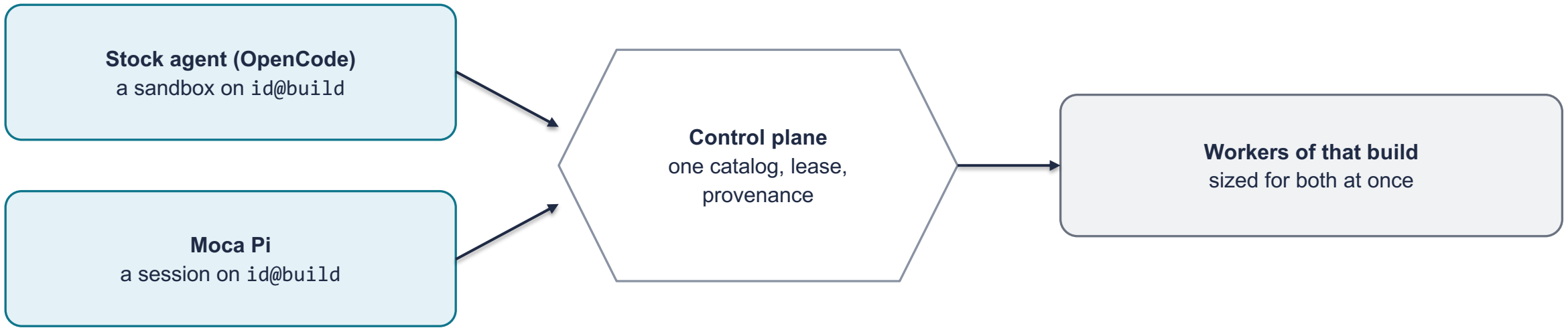
- **Self-serve, in three groups:** each group created, built and registered by policy, used on pinned builds, then retired; the host agent's removal checked before the next group.
- **Provenance:** every chain before an unchanged prefix after; an independent composer joined every link of every rep uniquely.
- **One instance left out:** its id is a template the deployment declares, which blocks a self-serve create under that id (G34).

Two agents on one template at once
the one-session minimum budget refused some commands at admission (G36): clients now retry and exit 75; budgets sized for the work run at once

Removal keeps workspaces
for the janitor; without it they stay on the host's disk

Two brains on one build, every link exact

The Moca Pi window, 2026-11-07: a stock agent and Moca Pi, a real model, the combined control plane



What the session now does	Seen
seeded from its build's own spec	`sandbox.seed.source: build`, the build = the pin
returns the workspace's patch with the turn	whole, its sha256 and size matching
leaves a `session` record at create and delete	its lease and build, on the template's chain
composes like a sandbox	every link of every rep unique, for both brains

What the windows found, and the fixes

Every finding has a register entry; Moca's are fixed and validated

Id	Found	Fixed in	Validated
G26	commands over 32 KiB hung the worker	the worker: refused at once	w2-w4
G25, G28	a registry host left the host in a reason	host agent redaction	w2-w4
G21, G29	duplicate builds; reuse of a removed snapshot	control plane reuse rules	w2-w4
G31	activation waited for the build poll	the wanted-state unit	w4
G19	a session could run on another build	the build label and filter	combined
G33	a rep's lease found only by time	sandbox records by digest	lifecycle
G35	a session got no seed, no patch, no record	seed from the build; patch; `session`	Moca Pi
G36	commands refused at a one-session budget	client retry; budget sizing	Moca Pi
G30	~20 s client stalls	the client's DNS (not Moca)	w4: 0 stalls

Since fixed
G34: an admin hands a declared template to a subject; G37: mocactl's template option takes `id@build`

The enhancements are open upstream, as reviewable PRs

rossoctl/moca, 2026-11-12: each stacked where it builds on another

Area	PRs	What it brings
Users	#762	enroll subjects and grant the admin role through the API
Templates	#760	a template store with builds, tags and readiness
	#766	self-serve create, gated by policy, with quotas
	#769	a subject's own gate record, for its own templates
	#759	a Moca Pi session can choose a sandbox template
Provenance	#756	template provenance in a hash chain, verifiable offline
mocactl	#767, #768	authoring commands; `--record`; token renewal
	#761, #755	retry a refused command (exit 75); `--version`
Hosts	#757, #758	microVM worker and snapshot fixes; deploy fixes
	#771, #772	from an image to a registered template; the host agent
	#773	a verify cache: a start re-hashes only changed snapshot files

Each one reviewed before its push
a tree check, a deploy-input diff, and a scan for hosts, ids, tokens and results

Next: land the PRs upstream, then the demo

In order, each with the user's decision first

1	**Upstream review**	answer the reviews of the open PRs; land them in stack order
2	**The next PRs**	activation, removal, memory, pacing and egress (#742-#745, #752), on the host agent (#772)
3	**The demo, not before 2026-11-19**	self-serve template and user lifecycle, live, on the tip the G50 window tested

Done since
synced with upstream main and tested on both namespaces; roles through the API on both; the client's whole path in mocactl

Users and templates both needed an operator; now both are API calls

Where the work started (template spec §1-§2; enrollment spec §1)

- **Templates:** each was a hand-run build, a hand-picked worker id, a settings edit and a redeploy; benchmarks need 30+ of them, built and retired by clients, not operators.
- **Users:** who may log in, and who is admin, were deployment settings; a change was a settings edit and a `setup.sh` run, and left no record.
- **One admin action took two cluster writes:** add a temporary admin by `setup.sh`, act through the API, remove the admin by `setup.sh` again.
- **And proof:** which image, which checks, which snapshot, and who changed whose access.

